Wastewater-Based Epidemiology for Emerging Pathogens in a Northern California Community Hospital, November 2024 to April 2025

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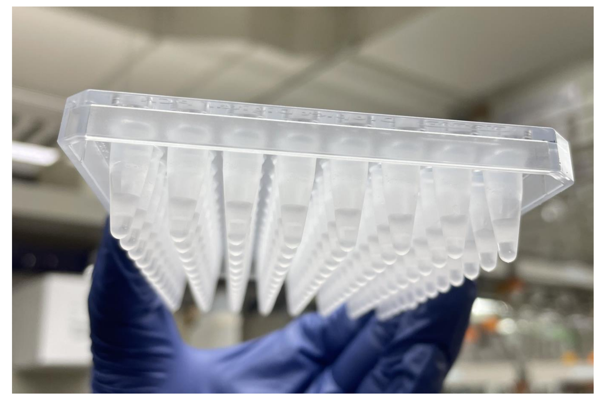
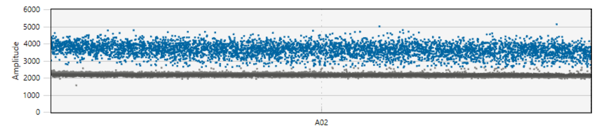
Background

- Wastewater-based epidemiology (WBE) has been effectively utilized to monitor respiratory viral circulation at the community level, its application in hospital settings remains largely unexplored
- We used WBE to gain insights into emerging pathogen circulation within a hospital environment

Methods

Weekly composite samples using autosampler → Nucleic acids extraction → Droplet digital PCR

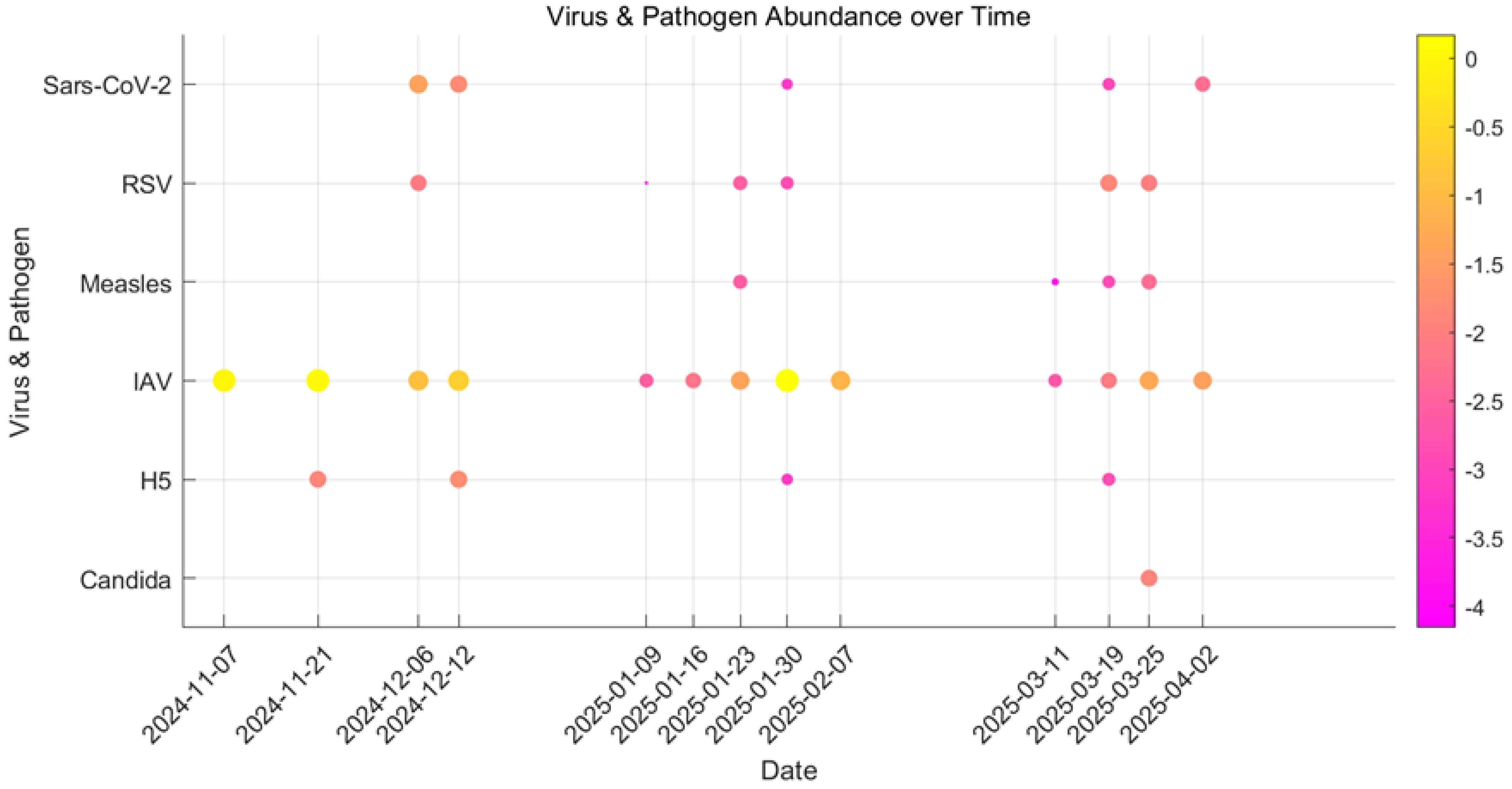




Conclusion

- WBE was reflective of classical epidemiology for SARS-CoV-2, RSV and Influenza A
- The detection of Measles and Influenza A subtype H5 despite lack of clinical cases may signify this strategy has a high sensitivity and may yield viral shedding post vaccination (measles) or consumption of dairy products (Influenza A subtype H5)
- The detection of Candida auris in wastewater may represent undetected/colonized patients in the hospital
- Further studies are necessary to establish a correlation between clinical isolates and the detection of antimicrobial resistance genes in wastewater.

Results



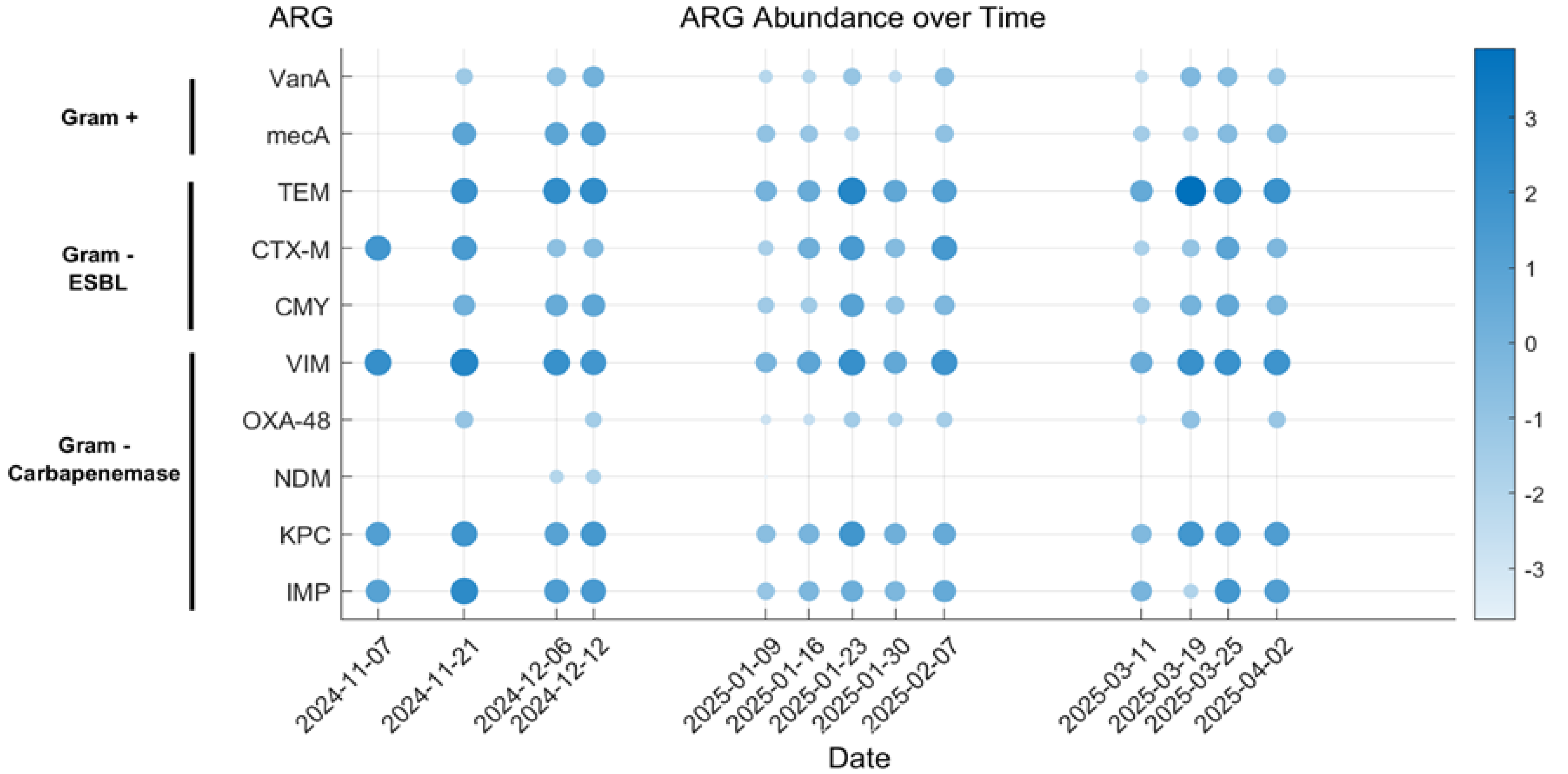


Figure 1. Virus and pathogen abundance plotted over time.
Figure 2. Antimicrobial resistance gene (ARG) abundance plotted over time.